

AI Gym & Fitness Assistant

Final Project Report — Functional Academic Prototype / MVP

1. Project Overview

AI Gym & Fitness Assistant is a modular fitness-support system that combines workout assistance, diet guidance, habit tracking, performance analysis, gym recommendations, smart gym settings, and conversational motivation in one application. The final implementation uses a Python/Streamlit user dashboard, modular Python components, a FastAPI backend, and a separate Streamlit admin analytics dashboard.

2. Objectives

- Provide computer-vision workout assistance with squat detection, repetition counting, knee-angle calculation, and basic form feedback.
- Generate general BMI, calorie, and goal-based meal suggestions through a modular Diet Coach.
- Track workout and missed days, calculate consistency, and predict the next workout date.
- Provide performance scoring, weekly planning, gym/program recommendations, and motivational conversational support.
- Provide backend APIs and an admin analytics dashboard for monitoring fitness metrics and module status.

3. Implemented Modules

Module	Implemented functionality
AI Gym Trainer	OpenCV + MediaPipe pose detection, squat counting, knee-angle calculation and basic form feedback using the webcam.
AI Diet Coach	BMI calculation, general calorie suggestion and goal-based meal suggestions.
Fitness Habit Tracker	Workout/missed-day tracking, consistency percentage, habit status and next-workout prediction.
Virtual Gym Buddy	LLM-powered conversational fitness support using local Ollama with Gemma 3:1B for motivation, workout guidance and supportive responses.
Performance Analyzer	Weighted performance score from form, repetition quality and consistency, with level and feedback.
Gym Recommender & Planner	Goal/experience-based gym or program recommendations and weekly fitness planning.
Smart Gym Assistant	Goal/experience-based intensity, resistance and rest-time settings, plus MQTT-based IoT integration using a simulated gym-equipment sensor environment.

4. Technology Stack & Integration

Application: Python, Streamlit. Computer Vision: OpenCV, MediaPipe. Data/Utilities: Pandas, NumPy. Visualization: Plotly. Backend: FastAPI and Uvicorn. Validation: Pydantic. Database: MongoDB. IoT messaging: MQTT using Mosquitto and Paho MQTT, with simulated equipment sensor data for the current academic prototype. Conversational AI: Ollama with Gemma 3:1B running locally. Scikit-learn is included in the environment, while current recommendation and habit logic is primarily Python/rule-based rather than trained predictive ML.

The FastAPI integration layer provides health, diet, habit, gym-recommendation/planning, smart-gym, performance and Gym Buddy services. The main user dashboard provides the user-facing interface. The Admin Dashboard displays performance, habit, goal-distribution and BMI analytics plus system-module status.

5. Testing and Results

- The AI Gym Trainer was tested with the webcam and successfully performed pose-based squat detection and repetition counting.
- The AI Diet Coach was tested with different dashboard goals, including Weight Loss and Muscle Gain, and produced expected goal-specific results.
- The Habit Tracker, Virtual Gym Buddy, Performance Analyzer, Gym Recommender & Planner, and Smart Gym Assistant were tested and produced expected outputs without reported runtime errors.
- The FastAPI backend was syntax-checked, imported successfully, launched with Uvicorn, and tested through its API documentation. Health and diet requests returned successful responses.
- The Admin Dashboard was launched separately and verified to display metrics, charts, user analytics, and module status.

- The final project includes source modules, requirements, README documentation, report, screenshots, and GitHub version control.

6. Deliverables Coverage

Required deliverable	Evidence / status
1. Fully functional AI-based Gym & Fitness Assistant	Working academic prototype integrating seven fitness modules.
2. Modular AI models	Modular workout computer vision, diet recommendation, habit/behavior analysis, conversational AI and fitness-support components. Several components are deterministic/rule-based rather than trained ML models.
3. Backend APIs and integration layer	FastAPI backend in api.py with health, diet, habit, gym-recommendation, smart-gym, performance and Gym Buddy endpoints.
4. Documentation and testing reports	README, final project report, screenshots and documented module/API testing are included.
5. Final deployment with admin dashboard and analytics	Local deployment demonstrated with the main dashboard, FastAPI service and separate Admin Dashboard with Plotly analytics. This is an academic/local deployment, not cloud production deployment.

7. Limitations & Future Scope

This is an academic prototype/MVP. Smart Gym equipment availability and sensor readings are currently demonstrated through an **MQTT-based simulated gym-equipment sensor environment**, rather than physical IoT devices. The MQTT architecture can be extended to physical gym sensors in future work. The Virtual Gym Buddy is now an **LLM-powered local chatbot using Ollama and Gemma 3:1B**. Cloud storage and cloud production deployment are not included in the current prototype. Diet and calorie outputs are general guidance and are not medical advice. Future work can add trained ML behavior models, physical IoT hardware integration, persistent cloud databases, cloud deployment, and a production frontend/backend architecture.

8. Conclusion

The final AI Gym & Fitness Assistant demonstrates the integration of computer vision, fitness recommendation logic, habit analysis, performance scoring, planning, LLM-based conversational support, MQTT-based simulated IoT integration, backend APIs, database support, and administrative analytics in a single academic system. Its modular structure provides a practical foundation for future development into a more advanced fitness platform.

Repository: github.com/adityakumarlenka15-byte/AI-Gym-Fitness-Assistant